Phase 1:

1A

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$
```

1B

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
```

Phase 2:

2A

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ gcc -o test_tree test_tree.o object.o tree.o index.o -lcrypto
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

2B

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ xxd .pes/objects/71/c5382acd7bd1fb1a618253220941fa6c56c956e89405dd0b5376ee235983bb | head -20
00000000: 7472 6565 2034 3500 3130 3036 3634 2061  tree 45.100664 a
00000010: 2e74 7874 002c f8d8 3d9e e295 43b3 4a87  .txt.,..=...C.J.
00000020: 7274 21fd ecb7 e3f3 a183 d337 6390 25de  rt!........7c.%.
00000030: 576d b9eb b4                             Wm...
```

Phase 3:

3A

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ ./pes status
Staged changes:
  staged:     a.txt
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_write.c
  untracked:  index.h
  untracked:  pes.h
  untracked:  tree.c
  untracked:  test_tree
  untracked:  tree.c.save
  untracked:  test_objects.c
  untracked:  test_write
  untracked:  pes.c
  untracked:  README.md
  untracked:  test_tree.c
  untracked:  Makefile
  untracked:  test_sequence.sh
  untracked:  test_objects
  untracked:  index.c
  untracked:  commit.h
  untracked:  commit.c
  untracked:  .gitignore
  untracked:  tree.h
  untracked:  object.c
```

3B

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ cat .pes/index
100664 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776690646 6 a.txt
100664 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776690774 6 file1.txt
100664 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776690774 6 file2.txt
```

Phase 4:

4A

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ ./pes log
commit 8b18f1fec3bfab15b76efb6ae7858e9198e78f68213175b8c721da1339c83738
Author: PES User <pes@localhost>
Date:   1776691109

    Third commit

commit 140173a8e7b03a7dfe55f2fd16f0c93d5a73e2b2d0bbd241420b6d91da0da41b
Author: PES User <pes@localhost>
Date:   1776691109

    Second commit

commit d6ab44c3693613b2b1d1a46efe125461020b5848fb8348feb5e35655b9ee035a
Author: PES User <pes@localhost>
Date:   1776691109

    Initial commit

commit 64a0fb4552460fa70ded2dd28946f12ec1e318091c2ca67f69b2be9a6c75abd9
Author: PES User <pes@localhost>
Date:   1776690676

    first067◆
```

4B

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/14/0173a8e7b03a7dfe55f2fd16f0c93d5a73e2b2d0bbd241420b6d91da0da41b
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/64/a0fb4552460fa70ded2dd28946f12ec1e318091c2ca67f69b2be9a6c75abd9
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/71/c5382acd7bd1fb1a618253220941fa6c56c956e89405dd0b5376ee235983bb
.pes/objects/86/6bb2c02d04ccc3bd4eb25ee3da18e5a87dcebd2b7435e71c65592d776fde86
.pes/objects/8b/18f1fec3bfab15b76efb6ae7858e9198e78f68213175b8c721da1339c83738
.pes/objects/ae/50d0a4c13cf6a472c4658e09f42e497f2c02e6b22820ac711a1f6b48ef262b
.pes/objects/b0/0e68191ee92e71a3e40627f58d82fecfc4a5d21186a4d27155a9e3fe5a1f20
.pes/objects/d6/ab44c3693613b2b1d1a46efe125461020b5848fb8348feb5e35655b9ee035a
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/refs/heads/main
```

4C

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ cat .pes/refs/heads/main
8b18f1fec3bfab15b76efb6ae7858e9198e78f68213175b8c721da1339c83738
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

Full Integration Test

```
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 97aa9e9d4d38... Initial commit

Log after first commit:
commit 97aa9e9d4d387afdc620b18416ab09c02bba3279c1e39dab573830057be74bf1
Author: PES User <pes@localhost>
Date:   1776691768

    Initial commit


--- Second Commit ---
Committed: ca6fb37ab5b3... Update file.txt
```

```
--- Third Commit ---
Committed: 2100f4f1eeca... Add farewell

--- Full History ---
commit 2100f4f1eecadd497e95c30d69a56312cb19835c6e43b7a4bc3fe8ba8ee94415
Author: PES User <pes@localhost>
Date:   1776691768

    Add farewell

commit ca6fb37ab5b3934fc52ee14fd774ce76f217944d98ca2444f82ecff1b9be7d86
Author: PES User <pes@localhost>
Date:   1776691768

    Update file.txt

commit 97aa9e9d4d387afdc620b18416ab09c02bba3279c1e39dab573830057be74bf1
Author: PES User <pes@localhost>
Date:   1776691768

    Initial commit
```

```
--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
2100f4f1eecadd497e95c30d69a56312cb19835c6e43b7a4bc3fe8ba8ee94415

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/21/00f4f1eecadd497e95c30d69a56312cb19835c6e43b7a4bc3fe8ba8ee94415
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/84/c5fed72c2dd9a3130e2e9b98033827f62793fb80d0295ccedf5fc7972d96eb
.pes/objects/97/aa9e9d4d387afdc620b18416ab09c02bba3279c1e39dab573830057be74bf1
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/ca/6fb37ab5b3934fc52ee14fd774ce76f217944d98ca2444f82ecff1b9be7d86
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/de/34fb483a5c16ce29ec3195dd505dfda7e35cc5fe16a99da4b2455b332a93cf
.pes/objects/ec/0af37b159ae92a5850ceaa6e497860729d6158aab2c6833b6068a1e3c5e8ef

=== All integration tests completed ===
nikhil@nikhil:~/PES2UG24CS317-pes-vcs$
```

Analysis Questions

5.1: Update .pes/HEAD to point to the new branch and replace the working directory with the target commit's tree; complexity arises from safely overwriting files and handling conflicts/untracked data.

5.2: Compare working directory files with index metadata (mtime/size or hash); if a tracked file differs from its indexed version and also differs between branches, block checkout.

5.3: Commits are created without a branch reference (dangling); they can be recovered by using the commit hash and creating a new branch pointing to it.

6.1: Traverse from all branch heads, mark reachable commits/trees/blobs using a hash set, then delete unmarked objects; roughly all objects reachable from ~100k commits (300k–500k objects) must be visited.

6.2: GC may delete objects that a concurrent commit hasn't yet referenced (race condition); Git avoids this using locking, reachability checks, and grace periods before deletion.